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Heat stress is a condition that often occurs in plants exposed to high temperatures for a prolonged period of time. Heat stress is a major factor in limiting the rate of photosynthesis.

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Heat stress decreases the light-dependent reaction of photosynthesis.

Explain why this leads to a decrease in the **light-independent reaction**.

[2 marks]

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Another effect of heat stress is a decrease in the activity of the enzyme rubisco. A decrease in the activity of an enzyme means that the rate of the reaction it catalyses becomes slower.

A decrease in the activity of the enzyme rubisco would limit the rate of photosynthesis.

Explain why.

[2 marks]

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Where precisely is rubisco found in a cell?

[1 mark]

- Figure 1** and **Figure 2** show their results.

Temperature / °C	Rubisco activity / arbitrary units
0	0.0
10	0.1
20	0.5
30	2.0
40	5.0
50	9.0

The graph shows the temperature dependence of Rubisco activase activity. The y-axis represents 'Rubisco activase activity / arbitrary units' ranging from 0.0 to 0.5. The x-axis represents 'Temperature / °C' ranging from 0 to 60. The activity starts at 0 at 0°C, begins to rise around 10°C, reaches a maximum of approximately 0.44 at 30°C, and then decreases sharply to about 0.05 at 50°C, remaining low up to 60°C.

Temperature / °C	Rubisco activase activity / arbitrary units
0	0.00
10	0.00
20	0.20
30	0.44
40	0.20
50	0.05
60	0.02

The scientists concluded that heat stress reduces the activity of rubisco in plant leaves by affecting rubisco activase.

Use all the information to evaluate their conclusion.

[4 marks]

[illegible]

